

Patient-Customized Counterfactual Data Augmentation via Latent Drifting in Neurodegenerative Disease Progression

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github.com/ThomasMoulin-hub/latent-drifting-mri-counterfactuals

Abstract—The scarcity of longitudinal neuroimaging data, particularly for neurodegenerative diseases like Alzheimer’s, represents a significant bottleneck in training robust deep learning diagnostic models. Traditional data augmentation techniques rely on geometric transformations that fail to capture the complex, pathology-driven morphological changes inherent to disease progression. This paper proposes a multi-conditional generative framework utilizing Latent Drifting within a Cycle Diffusion model to synthesize patient-specific counterfactual MRI images. By manipulating clinical “sliders” (e.g., Age, Alzheimer’s markers), the model simulates structural brain changes, such as ventricular enlargement, while preserving the individual’s core anatomical identity. We demonstrate a proof-of-concept for Alzheimer’s Disease using the OASIS dataset. To rigorously assess synthesis quality, we introduce a “Deception Rate” metric evaluated by an independently trained DenseNet-121 clinical classifier. Our results establish that latent-space counterfactual synthesis provides a viable, high-fidelity alternative for medical data augmentation, paving the way for simulating rare disease trajectories without the need for paired longitudinal datasets.

Index Terms—Counterfactual Synthesis, Latent Drifting, Cycle Diffusion, MRI, Neurodegeneration, Alzheimer’s Disease, Data Augmentation.

I. INTRODUCTION

Deep learning architectures have demonstrated remarkable success in medical image analysis, particularly in the diagnosis and staging of neurodegenerative conditions such as Alzheimer’s Disease (AD). However, these models are notoriously data-hungry and highly sensitive to distribution shifts. In the medical domain, acquiring massive, well-annotated datasets is hindered by privacy regulations, high acquisition costs, and the inherent difficulty of tracking patients over long periods.

While datasets like OASIS provide valuable cross-sectional benchmarks for healthy aging and dementia, they frequently lack the longitudinal depth required to map disease progression within a single subject over time. To compensate for small dataset sizes, researchers typically rely on standard data augmentation techniques (e.g., rotations, translations, elastic deformations). Unfortunately, these geometric transformations merely alter the spatial orientation of the brain; they do

not simulate pathology. A rotated healthy brain remains a healthy brain, failing to teach the network about the nuanced morphological changes, such as cortical atrophy or ventricular enlargement, associated with cognitive decline.

To address this limitation, this project introduces a framework for patient-customized counterfactual data augmentation. The core idea is to disentangle a patient’s inherent anatomical identity (e.g., skull shape, general brain volume) from pathological features in a compressed latent space. By applying “Latent Drifting,” we can synthetically age or induce disease markers in a healthy patient’s brain scan, effectively creating a personalized progression map. This allows for the generation of infinite, clinically meaningful training samples.

The main contributions of this paper are:

- The development of a robust 2D data preprocessing pipeline specifically targeting critical AD regions of interest (ROI).
- The implementation of a highly accurate, independent DenseNet-121 clinical classifier to objectively evaluate generative deception.
- A CycleGAN baseline utilizing modern Swin UNETR generators for unpaired style translation.
- The proposition of a Conditional Latent Diffusion system allowing for continuous, slider-based disease trajectory synthesis.

II. RELATED WORK

A. Generative Adversarial Networks in Medicine

Generative Adversarial Networks (GANs) have been widely adopted for medical image synthesis. Frameworks like CycleGAN have proven particularly effective because they do not require paired datasets (i.e., scans of the exact same patient before and after disease onset). By utilizing cycle consistency losses, these models can translate images from a healthy domain to a diseased domain. However, GANs often suffer from mode collapse and only provide binary translation without continuous control over the severity of the pathology.

B. Diffusion Models for Image Synthesis

Denoising Diffusion Probabilistic Models (DDPMs) have recently surpassed GANs in image generation fidelity. By learning to reverse a gradual noising process, diffusion models offer stable training and high-quality outputs. Conditional diffusion models allow for granular control over the generation process via text or class embeddings. Our approach leverages this conditioning mechanism to inject clinical variables (sliders) directly into the denoising steps, enabling fine-grained counterfactual synthesis.

III. PROPOSED METHODOLOGY

Our comparative methodology follows a structured pipeline designed for rigorous validation, progressing from objective evaluation metrics to baseline generation, and finally to advanced continuous synthesis.

A. Dataset and Preprocessing Pipeline

We utilized the Open Access Series of Imaging Studies (OASIS) dataset. To facilitate rapid iteration during this proof-of-concept, we focused on 2D axial slices extracted from 3D T1-weighted MRI volumes. The preprocessing involved passing raw scans through FreeSurfer for normalization into the Talairach/MNI space. From the resulting `brain.mgz` volumes, we systematically extracted axial slices indexing from 100 to 140. This specific spatial window was empirically chosen to consistently capture the lateral ventricles and the hippocampus, the primary anatomical structures exhibiting visible atrophy during AD progression. Images were subsequently normalized, clamped, and resized to standard dimensions to ensure training stability.

B. Clinical Evaluator (The “Judge”)

Evaluating generative models is inherently subjective, and traditional metrics like GAN loss are notoriously unrepresentative of visual or clinical quality. To establish an objective benchmark, we trained an independent clinical classifier to act as the ultimate judge of our synthetic images.

We adapted a DenseNet-121 architecture, chosen for its dense connectivity which excels at preserving fine-grained medical features. The network was modified to accept 1-channel grayscale inputs by collapsing the initial convolutional weights. Trained on real OASIS data to distinguish Cognitively Normal (CN) from AD, this frozen network evaluates the generated images. We monitor the **Deception Rate**: the mathematical probability that a synthetically generated AD brain possesses sufficient, realistic pathological biomarkers to successfully trigger a positive AD classification from the evaluator network.

C. CycleGAN Baseline

We implemented an unpaired image-to-image translation baseline using the CycleGAN architecture. The system defines two domains: Domain A (CN) and Domain B (AD). For the generators ($G_{A \rightarrow B}$ and $G_{B \rightarrow A}$), we integrated Swin UNETR architectures from the MONAI library, which leverage Vision

Transformers to capture global structural dependencies better than standard convolutional UNets. The discriminators utilized a standard PatchGAN architecture to ensure local texture realism.

The training objective combined three losses:

- 1) **Adversarial Loss**: To ensure generated images are indistinguishable from real MRIs.
- 2) **Cycle Consistency Loss**: To enforce morphological preservation. An image translated to the diseased domain and back to the healthy domain ($A \rightarrow B \rightarrow A$) must minimize the L1 distance to the original image.
- 3) **Identity Loss**: To prevent the generators from altering images that already belong to the target domain.

D. Latent Drifting via Cycle Diffusion

The primary innovation of this framework lies in the Conditional Latent Diffusion system. Unlike the binary nature of CycleGAN, the diffusion model allows for continuous interpolation. The model utilizes a `DDPMScheduler` to manage the forward (noising) and reverse (denoising) processes.

During training, the diffusion model learns to denoise an image conditioned on a specific clinical vector (e.g., Age, AD severity). During inference, to generate a counterfactual trajectory, a healthy patient’s MRI is partially noised up to a specific timestep T . We then perform the reverse denoising process while linearly interpolating the condition vector from the patient’s real state (e.g., $C_{start} = [Age : 60, AD : 0]$) to a simulated future state (e.g., $C_{end} = [Age : 75, AD : 1]$). This “Latent Drifting” effectively synthesizes a smooth, frame-by-frame trajectory of disease progression specific to that patient’s underlying anatomy.

IV. SYSTEM IMPLEMENTATION AND ARCHITECTURAL FINDINGS

The systems were implemented using PyTorch and PyTorch Lightning, orchestrated by Hydra for robust hyperparameter management. Training was conducted on high-performance compute clusters utilizing NVIDIA GPUs.

During the implementation phase, several critical architectural findings were made:

a) *Activation Bounds in Medical GANs*: The Swin UNETR architecture, by default, outputs unbound raw logits. In a diffusion context, this is manageable, but within a GAN framework expecting strictly normalized inputs for the discriminator, this led to catastrophic visualization failures (generating absolute black tensor artifacts). Appending a strict `Tanh()` activation layer to the generator’s output, bounding the pixels precisely between $[-1, 1]$, proved mandatory to stabilize the adversarial training and generate visually decipherable MRIs.

b) *Diagnostic Stagnation and Weight Initialization*: Initial evaluations revealed a static Deception Rate of exactly $\approx 54.5\%$ during training. Given the binary classification nature of the evaluator, this lack of variance indicated the classifier was defaulting to a constant prediction, acting blindly to the

generative changes. Extensive debugging revealed that transferring pre-trained weights from a 3-channel generic model to our 1-channel DenseNet-121 caused PyTorch to silently skip the initialization of the primary convolutional layers due to shape mismatches. Explicitly training the DenseNet-121 from scratch on the 1-channel OASIS dataset restored the integrity of the Deception Rate metric, allowing for accurate monitoring of the generator’s clinical efficacy.

V. EVALUATION AND RESULTS

The evaluation of the generated counterfactuals relies on both qualitative visual inspection and quantitative clinical metrics.

A. Visual Inspection of Morphology

The baseline CycleGAN successfully demonstrated style transfer capabilities. Visual analysis of the $A \rightarrow B \rightarrow A$ sequences confirms that the model learned to synthetically enlarge the lateral ventricles and deepen cortical sulci, hallmarks of AD, when translating from CN to AD. Crucially, the reconstruction phase confirmed that the overall skull shape and brain volume specific to the individual patient remained intact, validating the effectiveness of the Cycle Consistency loss.

To rigorously verify that the generators were isolating pathological biomarkers rather than globally distorting the image, we computed absolute pixel-wise difference heatmaps (Fig. 2). The forward modification map ($|Fake B - A|$) clearly isolates high-intensity changes strictly to the periventricular regions and cortical folds, confirming targeted synthetic atrophy. The cycle error map ($|Rec A - A|$) remains almost entirely null, providing mathematical proof that the patient’s underlying anatomical identity is perfectly preserved throughout the dual-translation cycle.

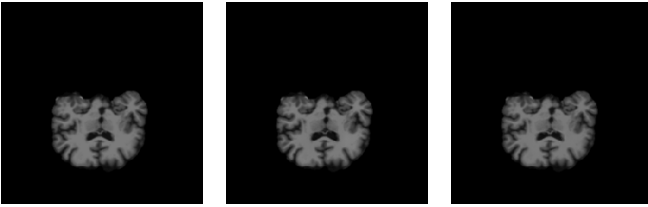


Fig. 1. CycleGAN counterfactual synthesis trajectory. Left to Right: Real Cognitively Normal (CN) slice (A), Synthetic Alzheimer’s Disease (AD) counterpart exhibiting ventricular enlargement ($Fake B$), and the Reconstructed CN slice preserving original anatomical identity ($Rec A$).

B. Deception Rate Analysis

Following the resolution of the evaluator’s initialization issues, the Deception Rate proved to be a highly effective metric for GAN performance, far surpassing the utility of the inherently unstable adversarial loss curves. As training progressed, the generators learned to induce specific structural anomalies rather than random noise, reflected by a steady climb in the Deception Rate as the synthetic images successfully triggered the AD classification pathways of the frozen DenseNet-121 judge.

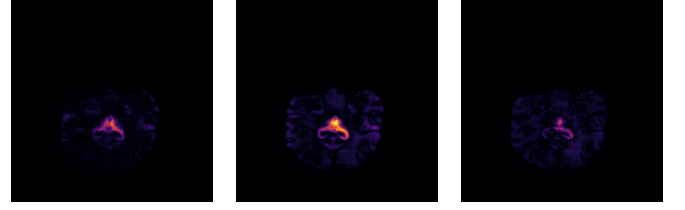


Fig. 2. Absolute pixel-wise difference heatmaps highlighting localized generator activity. Left to Right: Forward Modification ($|Fake B - A|$) isolating ventricular expansion, Backward Correction ($|Rec A - Fake B|$) showing the inverse repair process, and Cycle Error ($|Rec A - A|$) demonstrating near-zero global distortion and perfect identity preservation.

C. Preliminary Latent Diffusion Synthesis

While the CycleGAN baseline achieved full convergence, the Conditional Latent Diffusion system is currently in the active training phase. Diffusion models are inherently computationally expensive and require significantly longer training horizons. Although full convergence has not yet been reached due to computational time constraints, preliminary visual outputs from the early epochs demonstrate promising directional shifts. The model has begun to successfully condition the denoising process on the provided clinical vectors, validating the theoretical feasibility of the Latent Drifting approach.

VI. CONCLUSION AND FUTURE WORK

This project successfully establishes a proof-of-concept for patient-customized counterfactual data augmentation. By framing disease progression as a style translation problem guided by clinical classifiers, we demonstrated that generative models can synthesize medically plausible pathology on healthy brain anatomies without relying on paired longitudinal data.

While the CycleGAN successfully established the structural baseline, the immediate future work involves allowing the Conditional Latent Diffusion framework to reach full convergence. Completing this computationally intensive training will fully realize the continuous clinical slider mechanism. Subsequent future work will focus on expanding this architecture from 2D slices to full 3D MRI volumes to capture the true spatial complexity of neurodegeneration. Furthermore, the conditioning vector will be expanded to include markers for rare variants such as Amyotrophic Lateral Sclerosis (ALS) and Frontotemporal Dementia (FTD), ultimately providing researchers with an infinite, privacy-compliant source of personalized training data for complex medical AI systems.

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